Machine Learning 2 (APM_50441_EP/DSAIB - 2026-2027). PC7

Exercise 1 (Speed-up laws)

A program runs in time T_1 on one worker. A fraction f of its work can be parallelized perfectly, the rest is sequential.

1. *Amdahl's law.* Show that the speed-up with P workers is $S(P) = \frac{T_1}{T_P} = \frac{1}{(1-f)+f/P}$, and compute its limit when $P \to \infty$. Numerical application: $f = 0.95$, $P = 4$, 16, ∞.

2. *Gustafson's law.* In practice, a larger machine is used to solve a larger problem in the same time. If the parallel part grows with P while the sequential part stays the same, show that the *scaled* speed-up is $(1 - f) + fP$, where f is the parallel fraction of the time on P workers.

3. *Overheads.* Each worker adds a cost cP (starting the processes, copying the data to them...), so that $T_P = (1-f)T_1 + fT_1/P + cP$. Find the number of workers minimizing T_P. Explain the results of Lab 7, where 40 tiny tasks are *slower* in parallel than sequentially, while the 40 fits of the cross validation are faster, and why nested parallelism (parallel folds inside parallel models) oversubscribes the cores.

Exercise 2 (Complexity of cumulative sums)

We want to compute the cumulative sums $s_i = \sum_{k \le i} x_k$ of a vector of length n (Lab 7, Part 3).

1. Give the number of additions of the naive algorithm computing each s_i from scratch, and of the recursive one $s_i = s_{i-1} + x_i$.

2. In R, growing a vector with `res <- c(res, s)` copies it at each step. What is the total number of copied elements? Why does preallocating `res <- numeric(n)` solve the problem?

3. With about 1 ns per elementary operation in compiled code and about 100 times more in interpreted R code, estimate the running time of the naive and recursive algorithms for $n = 10^6$, in R and in C++ (`Rcpp`).

4. *Parallel prefix sum.* Describe a parallel algorithm computing all the s_i in $O(\log n)$ steps with $O(n)$ total work.

Exercise 3 (Map/Reduce for statistics)

A large dataset is split into partitions (e.g. the flights of each month, Lab 7). In a Map/Reduce computation, each partition computes a small summary (*map*), and the summaries are merged (*reduce*).

1. Give mergeable summaries for the mean and the variance: from $(n_A, \bar{x}_A, M_A)$ and $(n_B, \bar{x}_B, M_B)$, where $M = \sum_i (x_i - \bar{x})^2$, compute the summary of $A \cup B$.

2. Show that the least squares estimate can be computed exactly from the summaries $(\mathbb{X}_k^\top \mathbb{X}_k, \mathbb{X}_k^\top \mathbf{Y}_k)$ of the partitions k. What is the size of these summaries? Why is a single map/reduce pass not enough for the logistic regression? Why is the median not mergeable?

Exercise 4 (Back-of-the-envelope estimates)

The US flights dataset of Lab 7 has 5.8 million rows and 31 columns. Use the orders of magnitude of the lecture: reading 1 MB sequentially takes about $50\,\mu$s from RAM, $100\,\mu$s from an NVMe SSD and 4 ms from a hard disk.

1. Estimate the memory size of the table if all its columns are stored as 8-byte numbers. How long does it take to read it from RAM, from an SSD, from a hard disk? Reading the compressed CSV file actually takes several seconds: what is the bottleneck?

2. A query only uses 3 of the 31 columns. Explain why a columnar format (Parquet, Arrow, DuckDB) reads about 10 times less data than a row format, and what *predicate pushdown* adds.

3. When is it worth climbing the scaling ladder of the lecture (one core, several cores, a larger machine, a cluster, the cloud)? Compare the price of renting a machine with 64 cores and 512 GB of RAM for one hour (a few dollars) with the cost of an engineer's day.

Exercise 5 (Reservoir sampling (optional))

A stream of N items arrives one at a time, N being unknown in advance, and we want a uniform sample of k of them with $O(k)$ memory. Algorithm: keep the first k items; then, for the t-th item ($t > k$), draw j uniformly in $\{1, \dots, t\}$ and, if $j \leq k$, replace the j-th item of the reservoir by the new one.

1. Prove by induction on $t \geq k$ that after t items, each of them is in the reservoir with probability k/t.
2. Why is sampling a good first step for an analysis of a large dataset (the *sampling trick* of the lecture)? How does the precision of an estimated mean depend on the sample size k and on the size N of the dataset?

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